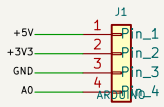
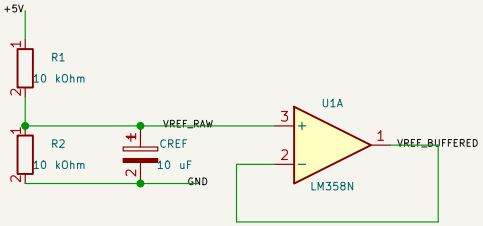
ED-4 QRNG PERFBOARD PROTOTYPE

Buffered reference, reverse-biased LED source and AC-coupled gain stage

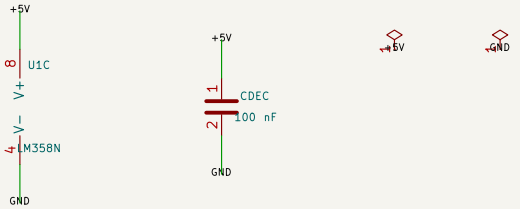
ARDUINO INTERFACE



MIDPOINT REFERENCE AND BUFFER

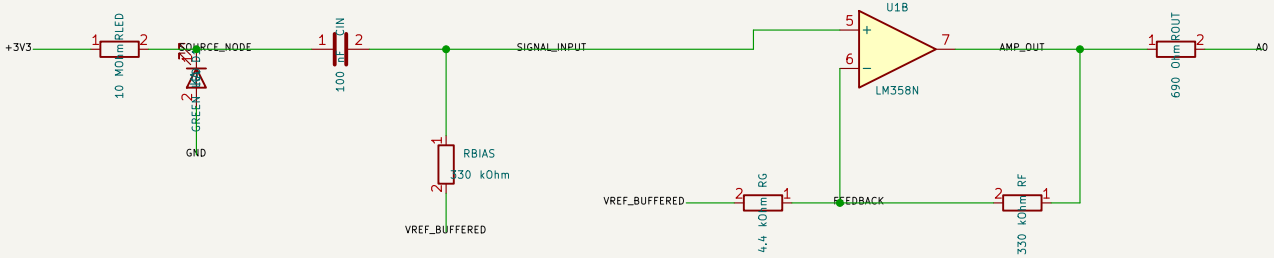


OP AMP POWER

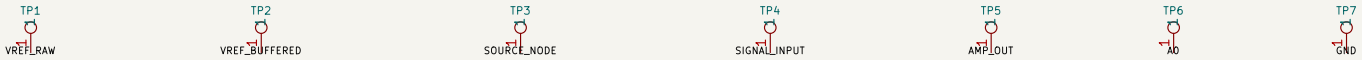


LED ENTROPY SOURCE AND AC COUPLING

SIGNAL GAIN AND ADC OUTPUT



BRING-UP TEST POINTS



Nominal U1B gain about VREF: 1 + 330 kOhm / 4.4 kOhm = 76.

D1 cathode connects to SOURCE_NODE; its anode connects to GND.

Arduino AREF is intentionally disconnected. Verify rails before connecting A0.

Arduino AREF remains disconnected
Verify component values and staged measurements before assembly
Connected engineering schematic for the ED-4 perfboard layout

Trishan Biswas

Sheet: /
File: ED4_Perfboard_Prototype.kicad_sch

Title: ED-4 QRNG Perfboard Prototype

Size: A4	Date: 2026-10-22	**Rev: DRAFT 2**
KiCad E.D.A. 10.0.1		Id: 1/1